The Chinese University of Hong Kong

Department of Computer Science and Engineering

KTL2401 - Invariant Hand Gesture Representation Learning with Augmented Contrastive Learning

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1 Introduction

1.1 Background

Hand Gesture Recognition (HGR) has played vital role in driving fields such as virtual/augmented/mixed reality (VR/AR/MR) and human-machine interactions. In particular, HGR is frequently used to extract instructions or language conveyed through hand gestures. However, the task of gesture recognition typically involves a number of challenges, including the possible variability in the dynamics of gestures, change in environment, occlusion, and the scarcity of high-quality well-annotated data.

Recent advancements in deep learning models and frameworks for keypoint extraction, such as MediaPipe [1] and DWPose [2], have set higher and higher expectations for gesture recognition systems, from recognizing limited sets of static gestures, to also distinguishing within a large variety of dynamic gestures often with only subtle differences.

To address issues like performance degradation due to disturbances and environment changes, many HGR frameworks also adopt multi-modal approachs, specifically, by including keypoints as one of the input streams [3, 4].

However, most existing frameworks focus only on training with a fixed set of gestures, which may not be sufficient for applications requiring custom gestures, or when high-quality annotated datasets are unavailable or too expensive to obtain. Specifically, traditional frameworks may not generalize well for different orientations of the gestrues that have relatively few samples within the dataset.

Meanwhile, semi-supervised learning emerges as a promising strategy with its ability to leverage limited sets of annotated data alongside larger sets of unannotated samples - which have lower difficulty to obtain compared to fully-annotated ones.

In this project, we would like to develop a method that maintains effectiveness and accuracy, even in scenarios where only a limited number of annotated reference samples is available, with contrastive learning approaches.

1.2 Motivation

At the same time, we also aim to explore the possibilities that semi-supervised and contrastive learning approaches could enable the models to learn a more structured and general representation of various hand gestures.

On one hand, existing computer vision models are sophisticated learners that are capable of learning complex correlations between concepts and objects, as shown in their success on significant benchmarks like ImageNet and WIT (Wikipedia-based Image Text Dataset) [5].

On the other hand, the constant failures in generating stable, correct hand structures in generative models has been demonstrating that the existing computer vision models still fail to understand comprehensively the structure of the hands and this is still an active problem in the field, as investigated in multiple works like HanDiffuser [6].

1.3 Non-Trivialness of the Problem.

In this project, we mainly rely on extracted keypoint landmarks instead of raw image frames for hand gesture recognition. Although it might seem that sign and hand gesture recognition based solely on keypoint landmarks is trivial, it can remain challenging for naive distance-based methods to discriminate between hand shapes that only differ by subtle change in the amount of bend of some finger.

For example, a simple nearest-neighbour classifier on raw landmark coordinates will often mislabel gestures that have higher tolerance to variations of finger positions (such as the ”hand open” gestures, like the gesture G8 in Figure 1), in which the accumulated small positional shifts from each finger can produce shifts in the phase space that is comparable to the level of shifts between two distinct signs (For example, gestures G4 and G8 in Figure 1).

Similar shifts in the phase space can also be produced under slight rotations or minor perturbations of individual joints, which can bring distinct hand configurations into near overlap in the landmark feature space. Meanwhile, depth estimations in hand landmark detection can be inaccurate and incon-

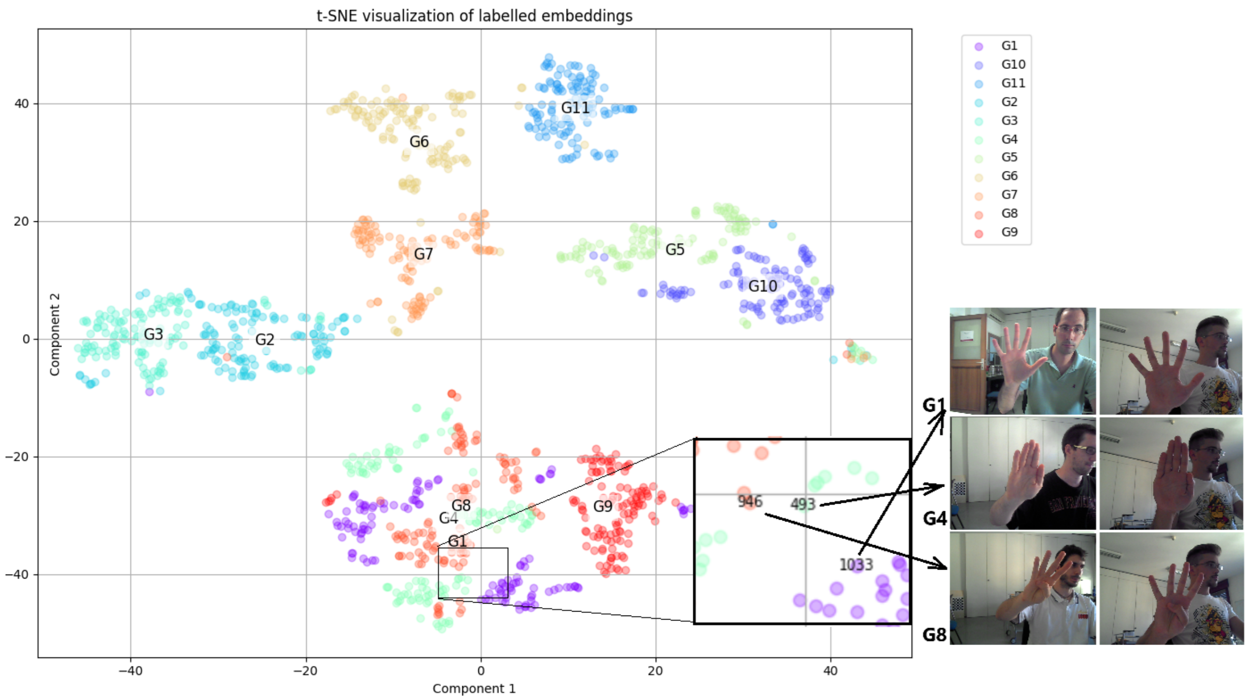


Figure 1: t-SNE plot for raw landmark over the Senz3d dataset

sistent.

1.4 Application

At the end of this project, we aim to produce a model that can enable quick adaptation to various downstream tasks, such as human-computer interaction and sign language, with minimal re-training or fine-tuning if possible.

Throughout the project, we will explore the effectiveness of various designs that we hypothesize would enhance the generalization capability of the models. Specifically, we would investigate:

1. Whether contrastive learning produces latent space that have better understanding on hand structure compared to supervised methods, and

2. Whether our approach can leverage a large unlabelled dataset for contrastive learning to minimize fine-tuning efforts, particularly when working with a relatively small gesture dataset.

2 Related Work

2.1 Contrastive Learning Approaches

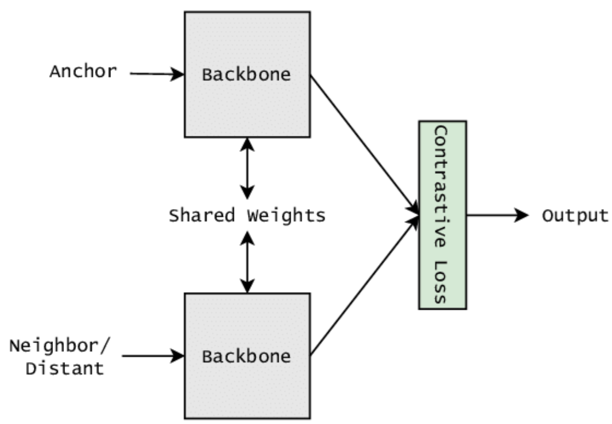


Figure 2: Architecture of a Siamese Network, with Contrastive Loss [7, 8].

In static image recognition, contrastive learning techniques, in particular, Siamese Networks [9], has played a crucial role under unsupervised and semi-supervised settings by utilizing shared parameters among the encoders for contrastive pairs, as illustrated in Figure 2, and have demonstrated its value in multiple fields including face identification. The architecture has contributed to the development of later frameworks such as SimCLR [10], MOCO [11], BYOL [12], etc., and have shown success on utilizing large sets of unannotated data by generating positive samples via data augmentation.

InfoNCE Loss. The InfoNCE loss is a well-known loss in the field that maximally utilizes the data within a batch of augmented duplets $(x_i, x_i^+), i \in [1..B]$ with the latent $(z_i, z_i^+) = (f(x_i), f(x_i^+))$, with the following formulation: [13, 14, 15]

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E}_{i\in I}\left[\log \frac{\exp\left(\text{sim}(z_i, z_i^+)\right)}{\exp\left(\text{sim}(z_i, z_i^+)\right) + \sum_{j\in I\setminus\{i\}} \exp\left(\text{sim}(z_j, z_i^+)\right)}\right]$$

where $\text{sim}(x, y) = (x \cdot y)/\tau$ for temperature τ and $f(x) = \text{Proj}(\text{Enc}(x))$ for projector network $\text{Proj}(\cdot)$ and encoder network $\text{Enc}(\cdot)$.

For the contrastive models trained with InfoNCE loss, *cosine similarity* is instead used to compute the similarity between any two latent feature vectors.

Supervised Contrastive Loss (SupCon). The supervised contrastive loss [16] utilizes also the information from the data from the same class apart from the augmentations, utilizing more comprehensively the information from the labelled datasets while preserving the properties of contrastive learning. It is given by:

$$\mathcal{L}_{\text{supcon}} = -\mathbb{E}_{i\in I}\mathbb{E}_{p\neq i:y(p)=y(i)}\left[\log \frac{\exp\left(\text{sim}(z_i, z_p)\right)}{\sum_{a\neq i} \exp\left(\text{sim}(z_i, z_a)\right)}\right]$$

Where, the supervised contrastive loss utilizes all augmented versions of all positive samples with the same class $y(p) = y(i)$.

2.2 Skeleton-Based Approaches

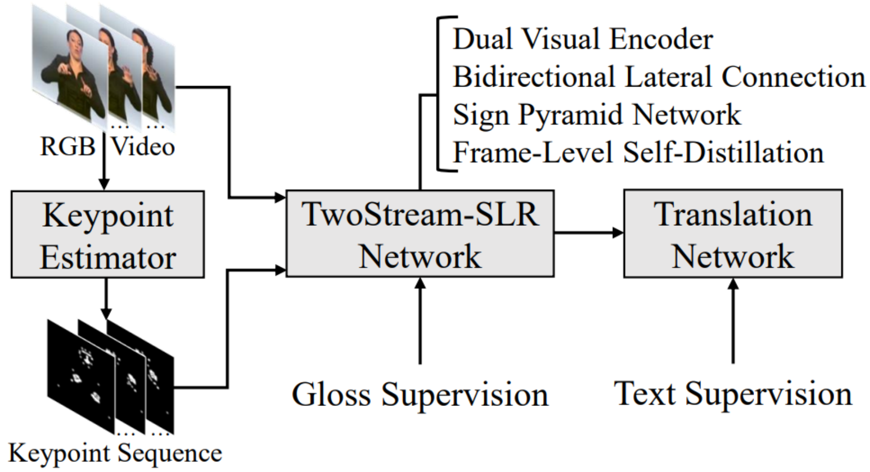


Figure 3: Architecture of TwoStreamNetwork [3].

With sophisticated pre-trained pipelines for extracting keypoints of hand and body gestures from videos, such as DWPose [2] and MediaPipe [1], many ”skeleton-based” HGR models, like Sung et al. [17], TwoStreamSLR [3], and STGCN-GR [18], are integrating keypoints as a new input stream (See Image 3). This integration can theoretically improve robustness against environmental variations, given a reliable pre-trained keypoint extraction model like MediaPipe. Additionally, a wider variety of augmentations, such as translation and rotation of individual joints, are easier to implement compared to using the video stream directly.

2.3 Dynamic Gesture Recognition

While HGR models for static gesture recognition have become sophisticated, dynamic gesture recognition has remained a challenging task, especially over a large number of classes. Methods like TwoStreamNetwork [3] and SlowFastSign [19] perform convolution also across the time dimension to integrate temporal data and has still been in the leading position over the task of sign language recognition.

Contrastive Learning over Temporal Data. Techniques such as Contrastive Predictive Coding [15] and Temporal Contrastive Learning [20], e2eET [4] have been discussed in the literature, which utilize different parts of the same clip and different clips as the positive and negative samples.

2.4 Curriculum Learning

While augmentations are often beneficial and have been widely used for training models that are robust to perturbations, overly aggressive or complicated augmentations introduced too early may hinder the efficiency of training. Curriculum learning addresses this problem by training with increasing complexity and difficulty over time [21].

3 Methodology

In the second part of our work, we focus on improving the generalization ability of the backbone model with more comprehensive augmentations and utilization of large unsupervised datasets, while also applying and extending similar techniques to dynamic datasets.

3.1 Datasets

Supervised Datasets. Extending on previous work, we trained and evaluated our models on both static and dynamic datasets. The full list of datasets used is listed as follows:

Dataset (* indicates imbalanced dataset)	**# Classes**	**# Samples**	**Image Size**
Static			
Synthetic ASL Alphabet (Lexset) [22]	27	27,000	512×512
Senz3D [23]	11	1,320	640×480
RWTH-PHOENIX-Weather 2014 MS Handshapes [24]	60*	3,000	92×132
Dynamic			
IPN Hand [25]	13	4,000 sequences	various

Table 1: Overview of static and dynamic hand-gesture datasets used in this project.

Static. Specifically, the RWTH-PHOENIX-Weather 2014 MS Handshapes Dataset [24] (Figure 4) is a largely unbalanced dataset featuring around 3,000 images in total, with over 60 different hand shapes. In this project, we are utilizing the diversity of the hand shapes within this dataset.

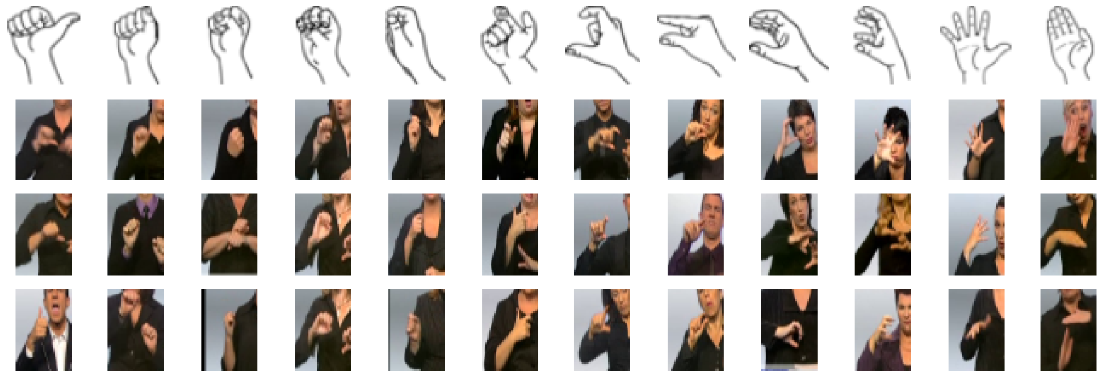


Figure 4: Sample images from the RWTH-PHOENIX-Weather 2014 MS Handshapes Dataset [24]

Dynamic. For the dynamic gesture recognition task, we trained and evaluated over the IPN Dataset [25], which is a dynamic gesture dataset consisting of over 4,000 gesture instances in total, with 13 static and dynamic gestures for interaction with touchless screens.

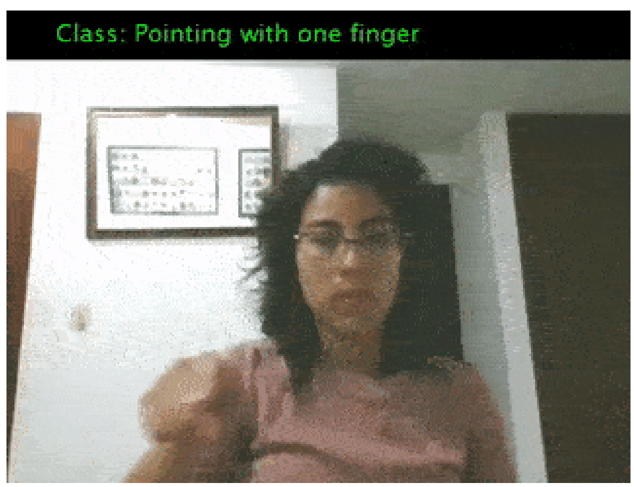


Figure 5: Sample images from the IPN hand dataset

Unsupervised Datasets. We also utilize the MANO hand model library [26] to generate a dataset of anatomically reasonable hand shapes by sampling and perturbing the low-dimensional parameter space.

The generated dataset consists of $G = 32$ augmentations for $N = 100,000$ distinct base hand shapes, which is then used for unsupervised portion of the training. The details of the mathematical formulation used for sampling in our implementation are provided in Appendix B.

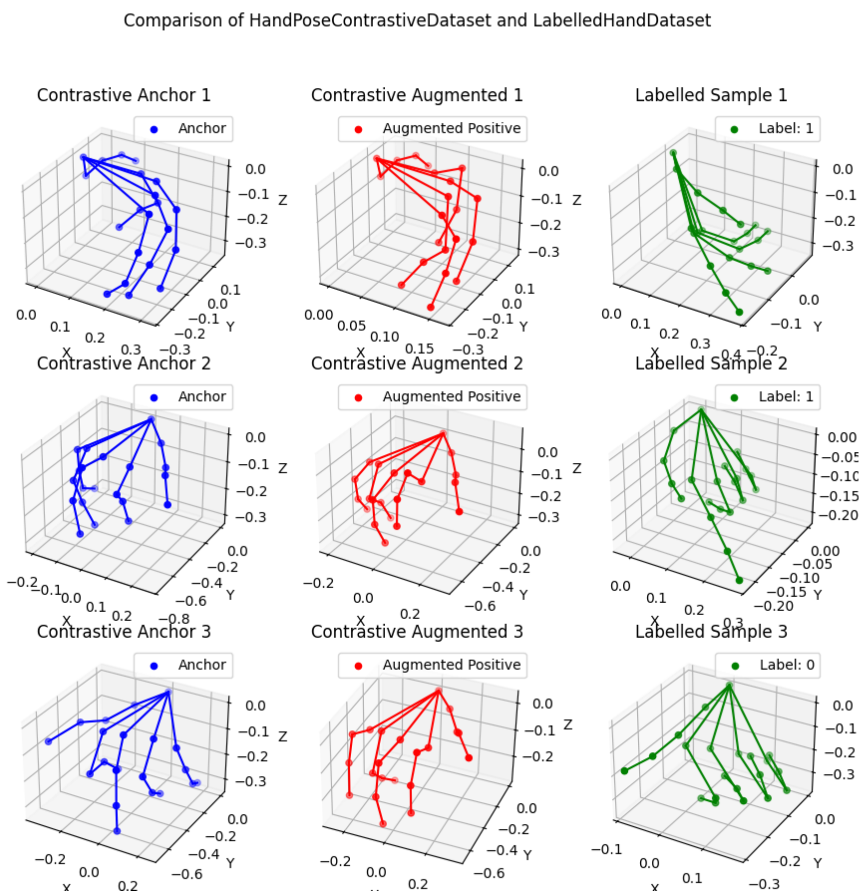


Figure 6: Comparison of MANO generated hands. Columns 1 and 2 show two augmented variants of the same base pose; Column 3 shows labelled samples from Lexset.

3.2 Problem Formulation

The gesture recognition task can be defined as a function denoted by $G : \mathbb{R}^{21\times 3} \to \mathbb{R}^{C}$, where C is the number of classes within the dataset.

For a general gesture recognition model, we expect that there exists a model

$M : \mathbb{R}^{21\times3} \to \mathbb{R}^{D+H}$, where the first D dimensions within the output features is approximately invariant to the rotation or offsets of the hand, and the later H dimensions captures the remaining information. ie we can decompose M into $M = [M_{\text{pose}}, M_{\text{rot}}]$ where M_{pose} extracts the rotational-invariant pose of the gesture, and M_{rot} extracts the pose-invariant information of the gesture, like rotations.

In this report, we focus on the extractor M_{pose} for the rotationally invariant features of the gesture.

3.3 Model Architecture

Static-Input Models. For the static datasets, we performed preliminary tests using shallower models before proceeding to more complex architectures, as the latter require significantly longer training time.

We explored three major base encoder types, including Multi-Layer Perceptron (MLP), Graph Convolutional Neural Network (GCN) [27] and Graph Attention Network (GAT) [28], where details will be covered in the next section.

We also explored the effects of input normalization (-I), utilization of angular parameterization (-A, or 6DoF) (as described in Appendix C) and pooling after the graph layers (-P).

3.4 Augmentations.

For training the static-input extractors, we train over both the supervised and unsupervised datasets with increasing augmentation magnitudes across epochs, inspired by the idea of curriculum learning.

By allowing the model to learn first from easier samples, then progressively increasing the difficulty of the task (i.e. the magnitude of the augmentation), we hope to improve the training stability and allow for a more efficient convergence [21].

In our formulation, we gradually increase the maximum angle of rotation θ applied on the inputs, where the rotation matrix R can be decomposed into the base rotation matrix R_b which is consistent across the same batch, and the child rotation matrices R_a that are restricted to smaller angles but are applied individually to each augmented versions of the input. The details of the augmentation is provided in Appendix D.

Also, in the augmentation process, a total of $G = 4$ augmented versions of the same input entry would be created, extending the batch size from B to $B \times G$ and providing $G - 1$ positive samples for each sample.

3.5 Loss Functions

In the process of training static-input feature extractors, we adopt the InfoNCE loss for the *unsupervised* dataset $\mathcal{D}_{\text{unsup}}$ and the SupCon loss for the *supervised* dataset $\mathcal{D}_{\text{sup}}$. For each epoch, the model is trained over an equal amount of batches from each dataset (where $\mathcal{B}_{\text{sup}} \subseteq \mathcal{D}_{\text{sup}}, \mathcal{B}_{\text{unsup}} \subseteq \mathcal{D}_{\text{unsup}}$) and the final loss is given by:

$$\mathcal{L}_{\text{total}}(\mathcal{B}_{\text{sup}}, \mathcal{B}_{\text{unsup}}) = \mathcal{L}_{\text{SupCon}}(\mathcal{B}_{\text{sup}}) + \mathcal{L}_{\text{InfoNCE}}(\mathcal{B}_{\text{unsup}})$$

4 Experiments

4.1 Base Model Choice

For the selection of the optimal base encoder, we trained the three types of base encoders only with the supervised dataset Lexset using SupCon, and evaluated their performances over the test set of Lexset using k-Nearest Neighbour. We intentionally included the graph-based networks because hand landmarks naturally form a skeletal graph, where each node is connected to its neighbours via bones, which we believe that this relation can be exploited by graph-based networks for them to learn more expressive and meaningful representation of the hand poses. The architecture of the models are listed in Table 2:

Model	**Total Layers**	**Graph Layers**	**MLP Head** (#Layers, Hidden Size)
MLP	4	N/A	4 layers, [256, 128, 64]
GCN	5	3 graph convolutional layers	2 layers, [64]
GAT	5	3 graph attention layers (4 heads)	2 layers, [64]
Embedding Size	128	**Dropout Rate**	0.3
Activation	Leaky ReLU (slope=-0.01)	**Batch Norm**	After every layer

Table 2: Overview of Model Architectures for Static Hand Gesture Feature Extraction

Method (%) (* implies evaluation with Cosine Similarity); Evaluated with k-NN	**Accuracy**	**F1 Score**	**Convergence**	**Notes**
Raw Landmarks	96.69	96.69	-	Baseline embedding
6DoF	93.22	93.20	-	Angular parameterized
MLP*	97.39	97.37	540ep	
GAT-A (6DoF)*	*98.04*	*98.04*	200ep	No pooling
GCN-A (6DoF)*	**98.45**	**98.45**	200ep	No pooling

Table 3: Evaluation Results using K-NN and Learned Embeddings

Among the three base encoders, Table 3 shows that GCN and GAT demonstrate better performance and faster convergence without signs of overfitting. Hence, the following experiments will mainly focus on these two models.

4.2 Curriculum Learning

To prevent strong data augmentations from hindering the initial training of the model, we implemented a curriculum learning strategy as described in Appendix D, and obtained the following results:

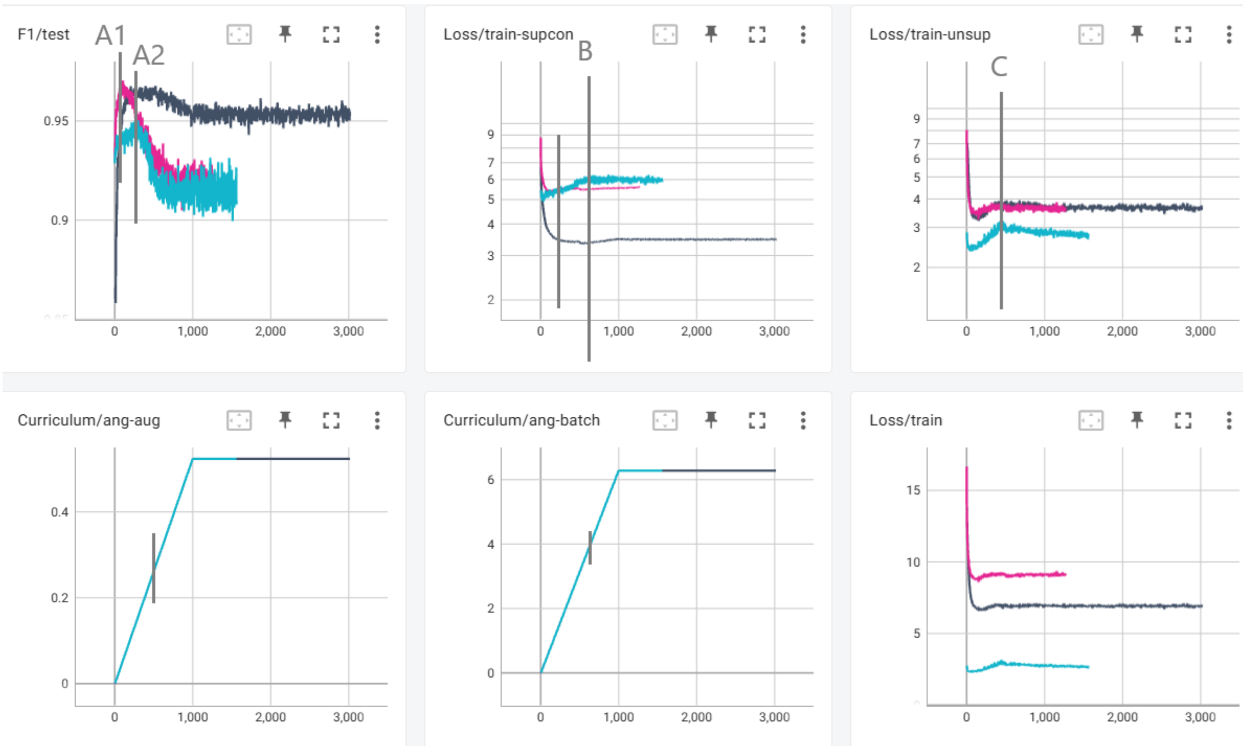


Figure 7: F1 score (top left), supervised (top middle) and unsupervised loss (top right) for GAT (grey, pink) and GCN (blue) under curriculum learning, and their maximum augmentation angles (bottom left and middle)

As shown in Figure 7, curriculum learning leads to an initial increase in accuracy and F1-score, reaching a peak before declining and leveling off around a fixed value. In contrast, the accuracies never exceeded 90% if the augmentations are fixed to its maximum value throughout the training.

Contrary to our hypothesis, our experiments indicate that the models may have insufficient capacity to effectively learn when the augmentation magnitudes are overly large. Specifically, the accuracies peaked at an intra-batch augmentation magnitude of approximately 0.3 and batch augmentation magnitude of 4.

Furthermore, while the supervised loss increases beyond point B and stablizes, the unsupervised loss continued to decrease after reaching a local peak at point C. This indicates that learning from the unsupervised dataset dominates when the augmentation magnitude is overly large.

Method	**Peak F1 (%)**	**Final F1 (%)**	**Handshape** Interclass (Mean ± Std)	**Senz3d**
No Model	96.69		0.829 ± 0.111	0.917 ± 0.038
		Intraclass:	0.888 ± 0.085	0.955 ± 0.015
No Curriculum	89.91			
GAT-A (6DoF, pink)	**97.04**	91.44	0.721 ± 0.165	0.471 ± 0.261
		Intraclass:	0.903 ± 0.039	0.938 ± 0.026
GAT-A (6DoF, grey)	*96.69*	95.46	0.824 ± 0.094	0.651 ± 0.156
		Intraclass:	0.930 ± 0.035	0.956 ± 0.020
GCN-A (6DoF, blue)	94.88	90.81	0.216 ± 0.421	0.043 ± 0.461
		Intraclass:	0.747 ± 0.124	0.872 ± 0.038

Table 4: Comparison of peak and final F1 scores along with interclass statistics for curriculum learning runs.

Table 4 also showed that the clusters formed within the latent space managed to improve from the baseline for the GAT models (where the separation between the class means increases, and the variances within the class distances reduces).

4.3 Dynamic Gesture Recognition

Model	**Input Dim**	**# LSTM Layers**	**Hidden Dim**	**Pooling**
LSTM-a	177	3	256	None
LSTM-windowed	177	3	256	AvgPool1d
LSTM-hierarchical	177	2 / 2 / 2 (body/hand/final)	256	None
LSTM-b	63	2		
RNN	63	2 (RNN)		

Table 5: Architectural details of the LSTM-based gesture models.

Model	**Input**	**Modality**	**Accuracy (%)**	**Convergence**
Baseline Results reported by IPN dataset official website [25]				
ResNeXt-101	32 frames	RGB-Flow	42.47	—
ResNet-50	32 frames	RGB-Flow	39.47	—
ResNeXt-101	32 frames	RGB-Seg	39.01	—
ResNet-50	32 frames	RGB-Seg	33.27	—
ResNeXt-101	32 frames	RGB	25.34	—
ResNet-50	32 frames	RGB	19.78	—
Our Work (Direct Training)				
LSTM-a	framewise	Landmarks	71.13	300 epochs
Bi-LSTM-a	framewise	Landmarks	80.61	300 epochs
LSTM-hierarchical	framewise	Landmarks	76.85	600 epochs
Bi-LSTM-hierachical	framewise	Landmarks	**84.78**	400 epochs
Our Work (Contrastive)				
LSTM-b (margin contrastive)	blockwise	Landmarks†	**94.02**	
RNN (margin contrastive)	blockwise	Landmarks†	76.81	
Bi-LSTM-windowed (SupCon)	32 frames	Landmarks	*83.75*	700 epochs

Table 6: Comparison of various architectures evaluated upon the IPN dataset.
† Indicates that only hand keypoints are used; Otherwise, full-body keypoints are used.
[2] Bi-LSTM indicates Bi-directional LSTM.

Data Preparation and Augmentation

Each sequence of dynamic gestures is rotated about one of x, y and z axes by -5 to 5 degrees, and enlarged or shrinked by 1.2 times or 0.8 times. Each sequence is also linearly interpolated and sampled with different amount of landmark points as to simulate different speed the gesture is performed. (between 0.75x and 1.5x speed) Movement of landmark 0 (wrist position) is used instead of the original wrist position as position of hand does not change the gesture classes.

Contrastive with LSTM.

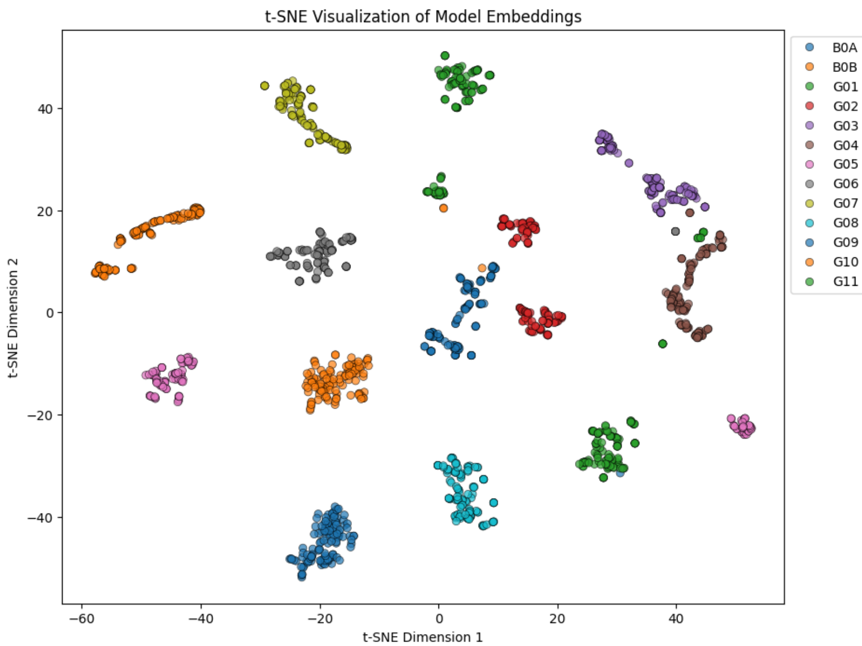
We trained a 2-layer LSTM model with contrastive loss to train on the IPN dataset using Mediapipe landmarks as input. The model consists of 2 layers of LSTM and 1 layer of linear layer. Test accuracy of 0.9402 is reached. Below shows the t-SNE graph.

RNN.

Similar to training the LSTM, RNN is also tested using the IPN data set. The model has 2 RNN layer with 1 linear layer. The results obtained are unstable test accuracies and the highest achieved is 0.7681.

Trail comparison.

By converting landmark coordinates from Mediapipe to latent space using the



model trained for static images, we can get the trail of the dynamic gesture. We then compare the similarity of any 2 samples by using metrics including DTW, average cosine distance and frechet distance. In most of the metrics, samples from the same class has mean less than the mean from different classes. 20bn-jester dataset is used.

Metric	Intra-class Mean (normalized)	Std	Inter-class Mean	Inter Std
DTW	1.0000	0.3649	1.2372	0.3416
Frechet	1.0000	0.3282	1.1905	0.3131
Avg. Euclidean	1.0000	0.4956	1.2082	0.4828
Avg. Cosine	1.0000	1.1631	1.3047	1.1988
Hausdorff	1.0000	0.3581	1.2064	0.3482
Correlation	1.0000	0.6871	1.4038	0.7355
Procrustes	1.0000	0.4392	1.1491	0.4077

Table 7: Intra-class and Inter-class Similarity Metrics

The metrics scores are used as input for a neural network of 3 fully connected layers with sigmoid activation at the end to predict if the 2 samples are from the same class. The accuracy obtained is 0.7234.

5 Discussions

In this report, while we have covered various contrastive approaches over both static and dynamic gestures, the original objective of building a general, unified, and invariant hand gesture representation model remains only partially fulfilled.

Our experiments demonstrate that the incorporation of contrastive learning significantly enhances the performance and stability of the model in constrained scenarios. However, the resulting models still lack full generalization to unseen gesture types, variations or spatiotemporal transformations.

5.1 Future Work

In the future, we expect later work can expand upon our work and to train a true, general hand gesture encoder that is able to capture the general rotation-invariant and scale-invariant information of gestures that can be used to quickly adapt to various downstream tasks including dynamic gesture recognition.

To be precise, we expect future work to build a single model that could recognize and capture both static and dynamic features from videos, that is robust against various orientation changes and perturbations.

A Division of Labour

To ensure clarity in our contributions, Table 8 outlines the division of labor between group members for key project tasks.

Task	TAM, Ka Ho	MAN, Tze Wa
Dataset Preparation	Generated and preprocessed unsupervised dataset using MANO hand model	Curated and preprocessed supervised datasets (Lexset, Senz3D, RWTH-PHOENIX, IPN Hand)
Curriculum Learning	Designed and implemented curriculum-based augmentation strategy for static models	-
Static-Input Models	Developed and tested MLP and GAT models, including input normalization and angular parameterization	-
Dynamic-Input Models	Designed and tested Bi-LSTM with SupCon and windowed approaches	Implemented RNN and margin contrastive LSTM models
Contrastive Learning	Configured unsupervised contrastive augmentations (InfoNCE, SupCon) for static models	Developed contrastive learning for sequences (margin contrastive) and DTW integration
Experimentation	Conducted experiments for static models (MLP, GCN, GAT) and analyzed F1 scores	Ran experiments for RNN and LSTM models, evaluated loss trends

Table 8: Division of Labor for Project Contributions

B MANO parameterization

In the parameter space for the MANO hand model, let

$$\beta_{\text{base}} \in \mathbb{R}^{10}, \quad p_{\text{base}} \in \mathbb{R}^{45}$$

denote the MANO perturbation and pose parameters. Define the index sets S_f, S_r and S_t which correspond to the list of indices of features controlling the forward bending, sideways bending (rotation) of each finger, and the bending of the thumb, respectively.

Then, we sample the base parameters for each entry as

$$\beta_{\text{base}} \sim \text{Uniform}(-\sigma_\beta,\ \sigma_\beta),\ \ p_{\text{base},i} \sim \mathcal{N}(-b_i,\ \sigma_{p,i}^2),$$

where

$$\sigma_\beta = 2.0,\ \ \sigma_{p,i} = \begin{cases} 2.0, & i \in S_f \cup S_t, \\ 0.3, & i \in S_r. \end{cases},\ \ b_i = \begin{cases} 1.5, & i \in S_t, \\ 0.5, & i \in S_f, \\ 0, & i \in S_r. \end{cases}$$

We generate G augmented samples from the base parameters by adding Gaussian noise:

$$\beta_{\text{aug}}^{(j)} = \beta_{\text{base}} + \varepsilon_\beta, \quad \varepsilon_\beta^{(j)} \sim \mathcal{N}(0,\ \tilde{\sigma}_\beta^2), \quad \tilde{\sigma}_\beta = 0.1,$$

$$p_{\text{aug},i}^{(j)} = p_{\text{base},i} + \varepsilon_{p,i}, \quad \varepsilon_{p,i}^{(j)} \sim \mathcal{N}(0,\ \tilde{\sigma}_{p,i}^2), \quad \tilde{\sigma}_{p,i} = 0.1\,\sigma_{p,i}.$$

Afterwards, the j-th generated hand shape is obtained via the MANO forward model

$$H^{(j)} = \mathcal{M}(\beta^{(j)}_{\text{aug}}, p^{(j)}_{\text{aug}}),$$

and is then normalized and rescaled so that the sampled hand shapes have coordinates approximately lie within the range $[-0.4, 0.4]$ and have the standard deviation of each coordinate axis $\sigma_x = \sigma_y = \sigma_z = 0.1$.

The generated dataset is of the shape $[N, G, 21, 3]$, where we chose the number of entries to be $N = 100,000$ and the number of augmentations per entry to be $G = 32$.

C Angular Parametrization

Implemented with Python library `scipy`, the relative quaternion orientations are computed from the difference in orientation compared to the previous node. The wrist node orientation is computed based on the facing direction of the palm, while the other node orientations are computed from the direction of the edge connecting to that node, so that the normal vector always point in the direction that the finger curls towards (See Figure 8).

This approach is adopted in hope of improving the model's understanding of relative rotations. The quaternion approach is adopted as it is known to be more robust in representing rotations while avoiding the typical Gimbal lock problem and inconsistency in rotation representation that would present in other methods like matrix representation or using Eurler angles [29].

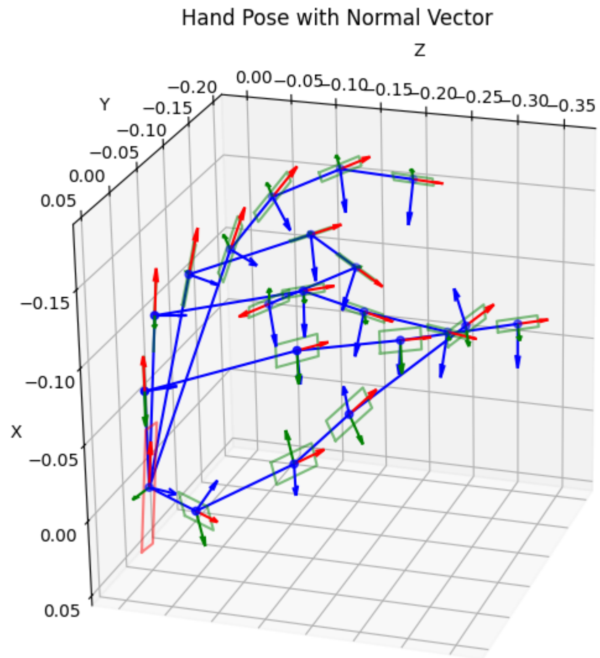


Figure 8: The axes corresponding to the "absolute orientation" of each hand landmark.

D Curriculum Scheduling for Static Datasets

Given grid size G, we augment the input batch into the shape of $[B, G, 21, 3]$. That is, we would have G augmentations for every entry within the batch. The augmentation of each batch is generated as follows:

- Base augmentation: $R_b(t) = \mathrm{RandomRotation}(\alpha_b \min\{t/T_b, 1\})$
- Child augmentations: $R_a^{(j)}(t) = \mathrm{RandomRotation}^{(j)}(\alpha_a \min\{t/T_a, 1\})$
- For each batch from the supervised dataset $\mathbf{L} \in \mathbb{R}^{B\times C}, \mathbf{B} \in \mathbb{R}^{B\times 21\times 3}$, the augmented sample have the shape $\mathbf{B}' \in \mathbb{R}^{B\times G\times 21\times 3}$, which is then flattened into the shape $(B \cdot G) \times 21 \times 3$.

    - Where, the j-th augmented sample of the i-th entry is given by: $\mathbf{B'}_{ij} = R_b(t)R_a^{(j)}(t)\mathbf{B}_i$

- For each batch $\mathbf{B} \in \mathbb{R}^{B\times G_D\times 21\times 3}$ from the unsupervised dataset that is generated with G_D augmentations, the augmented samples have the shape $\mathbf{B'} \in \mathbb{R}^{B\times G\times 21\times 3}$, which is then flattened into the shape $(B \cdot G) \times 21 \times 3$.
    - Where, the j-th augmented sample of the i-th entry is given by: $\mathbf{B'}_{ij} = R_b(t)R_a^{(j)}(t)\mathbf{B}_{iS_j}$ and $S \in \mathbb{R}^G$ is a random sample of size G from the range $[0, G_D)$.

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